# TQUK Functional Skills Qualification in Maths at Level 2

## Examination Past Paper 8

**Please complete the details below using black or blue ink. Use BLOCK CAPITALS.**

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

**Instructions:**

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

**Information:**

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

**Items:**

- you **will need** a pen with black or blue ink, a pencil, a ruler and an eraser (for diagrams, graphs and charts only)
- you **will need** a basic calculator for Section B only
- you **will need** a protractor
- you **will not** need any other stationery or equipment.

**Time allowed:**
**30 minutes** for Section A (Non-calculator)
**90 minutes** for Section B (Calculator)

**Do not open this examination paper until you are told to do so.**

**For examiner use only**

| | Marks available | Marks awarded | Second marks |
|---|---|---|---|
| Section A | 15 | | |
| Section B | 45 | | |
| Total marks | 60 | | |

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# Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

**1.** Work out 5.817 – 2.75

| | |
|---|---|
| | |
| **Answer** | |

**[1]**

**2.** Work out 4 × 3.145

| | |
|---|---|
| | |
| **Answer** | |

**[1]**

Past paper

**Questions continue on the following page**

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**3.** Convert 12 kilometres into miles.

*Use 1.6 kilometres = 1 mile*

| | |
|---|---|
| **Answer** | **miles** |

**[2]**

**4.** There are 824 students at a school.

112 of them are in their final year.

One student is chosen at random.

What is the probability that this student is in their final year?

Give your answer as a fraction in its simplest form.

| | |
|---|---|
| **Answer** | |

**[2]**

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**5.** Morgan is the manager of a concert arena.

The number of tickets sold in two years is shown:

| **Year 1**: | **Year 2**: |
|---|---|
| 784 693 tickets | 523 154 tickets |

Morgan thinks that the total number of tickets sold in these two years is more than 1 400 000

Is Morgan correct?

**Show how you decide**.

| | |
|---|---|
| | |
| **Answer** | |

**[2]**

Past paper

**Questions continue on the following page**

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**6.** Work out the original cost of this television:

| **Television** |
| --- |
| 20% off |
| Now only £240 |

| | |
| --- | --- |
| | |
| **Answer** | **£** |

**[2]**

**7.** Work out:

$8\frac{1}{4} + 6\frac{1}{5}$

Give your answer as a mixed number.

| | |
| --- | --- |
| | |
| **Answer** | |

**[2]**

Past paper

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**8.** Dua is going to make a gold bracelet using 50 cubic centimetres ($cm^3$) of liquid gold.

Dua knows the density of gold is 19.3 $g/cm^3$.

How many grams of gold will Dua need to make the bracelet?

*Use:* $\text{Density} = \frac{\text{mass}}{\text{volume}}$

| | |
|---|---|
| **Answer** | **grams** |

**[3]**

## End of Section A.

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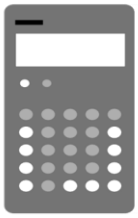
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**Section B begins on Page 7.**

Past paper

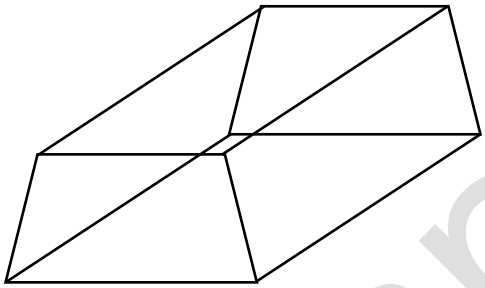
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## Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

**1.** What is this shape?



| | |
|---|---|
| **a)** Pyramid | **b)** Trapezium |
| **c)** Cuboid | **d)** Prism |

| | |
|---|---|
| **Answer** | |

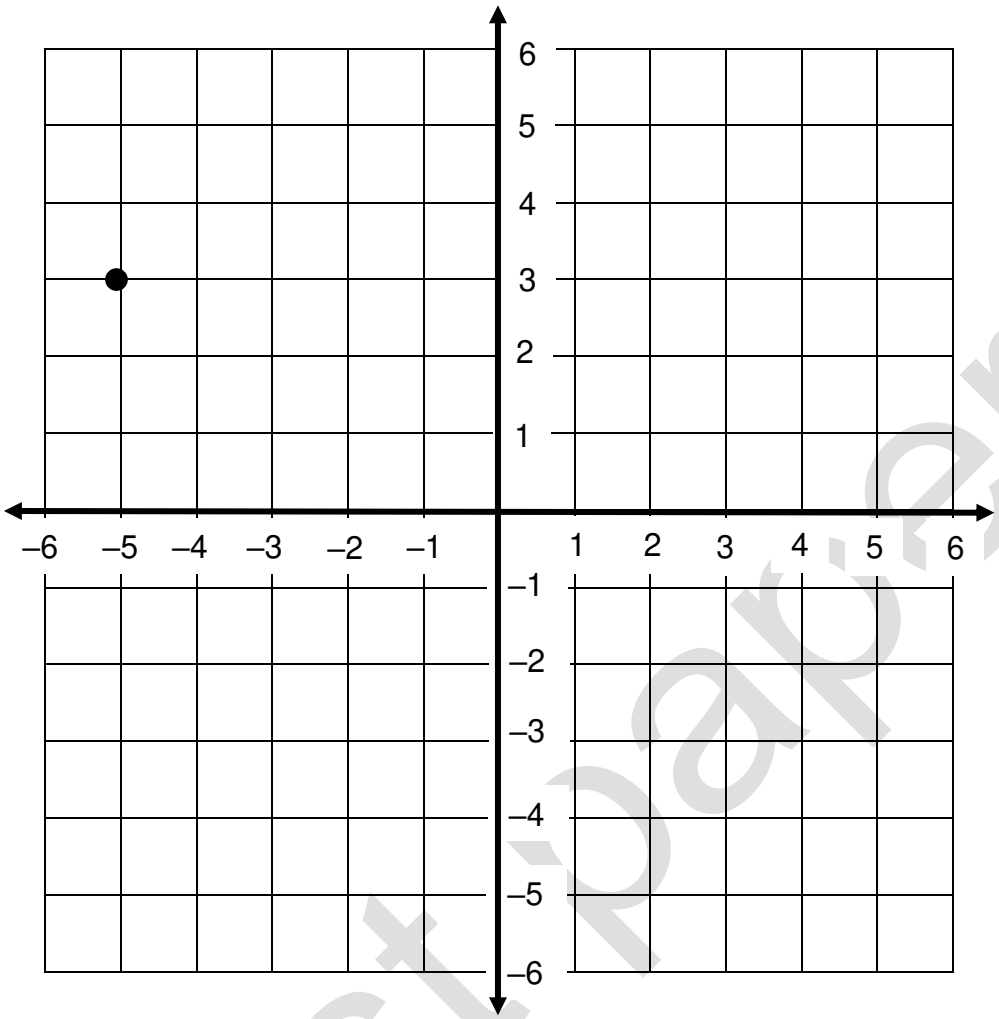
**[1]**

Past paper

**Questions continue on the following page**

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**2.** Write down the coordinates of the point plotted on the grid.




| **Answer** | ( , ) |
|---|---|

**[1]**

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**3.** Zola wants a new sofa.

Zola can choose either a grey or blue sofa.

Both sofas have the same original price.

| **Grey sofa** | **Blue sofa** |
|---|---|
| $\frac{1}{9}$ off | 12% off |

Which sofa has the bigger discount?

**Show how you decide.**

| | |
|---|---|
| | |
| **Answer** | |

**[2]**

Past paper

**Questions continue on the following page**

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**4.** Work out:

$$\frac{4^2 + 40 \div 2}{15 - 3 \times 4}$$

**Show your working**.

| | |
|---|---|
| | |
| **Answer** | |

**[2]**

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**5.** A local farm sells milk in glass bottles for £0.80 per pint.

The farm increases the price to £1.00 per pint.

Work out the percentage increase in price.

| | |
|---|---|
| **Answer** | **%** |

**[2]**

**6.** Tai needs their house painted within 10 days.

It will take 3 painters 6 days to paint the house.

Will 2 painters working at the same rate get the house painted within 10 days?

**Show how you decide.**

| | |
|---|---|
| **Answer** | |

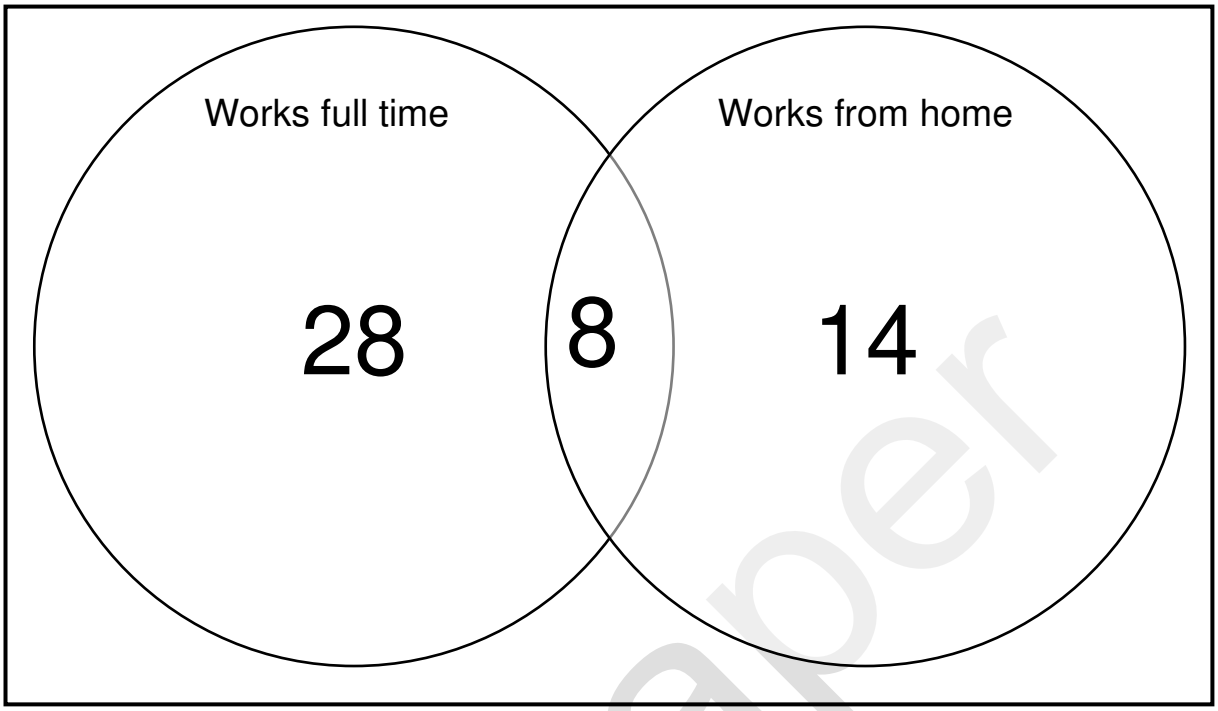
**[2]**

Past paper

**Questions continue on the following page**

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**7.** The Venn diagram shows the working patterns of a group of employees.




One of these employees is chosen at random.

What is the probability that this employee works full time?

Give your answer as a decimal **and** a percentage.

| | | |
|---|---|---|
| **Answer** | a) Decimal: | b) Percentage: |

**[3]**

Past paper

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**8.** Harper walks 15 kilometres in 200 minutes.

Harper thinks this is 4.2 kilometres per hour (kph).

Is Harper correct?

**Show how you decide.**

| | | |
|---|---|---|
| | **Answer** | **kph** |

**[3]**

Past paper

**Questions continue on the following page**

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**9.** Charu needs to estimate the curved surface area of a hemisphere.

Charu knows:

- the radius of the hemisphere is 9 cm
- $\pi = 3.14159$

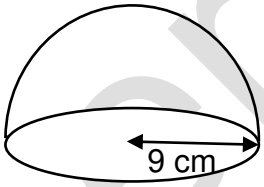
Charu rounds $\pi$ to the nearest whole number to estimate the curved surface area.

What answer should Charu get?

*Use:*

$$A = \frac{4\pi r^2}{2}$$

*Where:*
*A = curved surface area of sphere*
*r = radius of hemisphere*




Not drawn to scale

**Show your working.**

| | |
|---|---|
| **Answer** | **cm²** |

**[3]**

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**10.** Drew needs to order 200 paper cups for a party.

Drew needs small, medium and large cups in the ratio 14 : 4 : 2

Drew thinks there will be 108 more small cups than large cups.

Is Drew correct?

**Show how you decide.**

| | |
|---|---|
| | |
| **Answer** | |

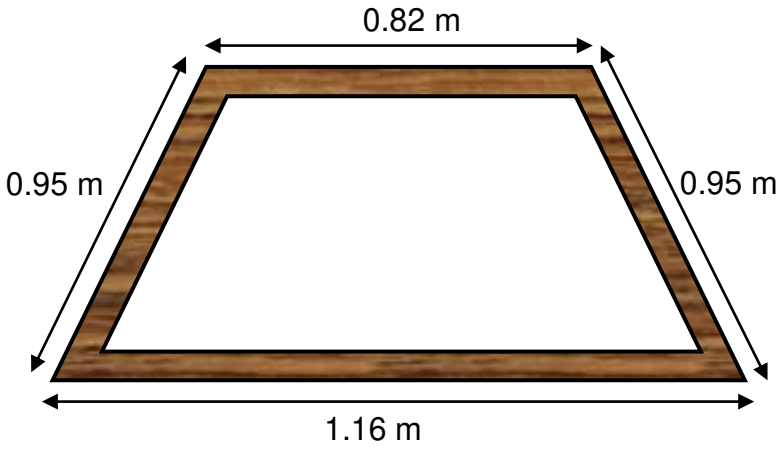
**[3]**

Past paper

**Questions continue on the following page**

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**11.** Kai wants to make a wooden picture frame with the dimensions shown below:




Not drawn to scale

Kai wants to buy 14% more wood than needed to allow for any mistakes.

Kai buys 4.5 metres of wood.

Has Kai bought enough wood?

**Show your working.**

| | |
|---|---|
| **Answer** | |

**[4]**

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**12.** Billie owns a café. Deepal is the café manager.

The table shows how many customers the café had over two weeks.

| | **Monday** | **Tuesday** | **Wednesday** | **Thursday** | **Friday** |
|---|---|---|---|---|---|
| **Week 1** | 22 | 38 | 46 | 30 | 44 |
| **Week 2** | 23 | 40 | 45 | 42 | 20 |

Deepal thinks that on average there were more customers per day on week 1.

Billie disagrees.

Show how **both** could be correct.

**Show your working.**

| |
|---|
| |
| **Give reason here:** |

**[4]**

Past paper

**Questions continue on the following page**

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**13.** Ashley wants to buy a motorbike for £9750

Ashley paid £5000 into a new savings account two years ago.

The savings account paid 4% compound interest per year.

Ashley will use all the money in the savings account as a deposit towards the motorbike.

The rest will be paid in 40 equal monthly payments.

How much will each monthly payment be?

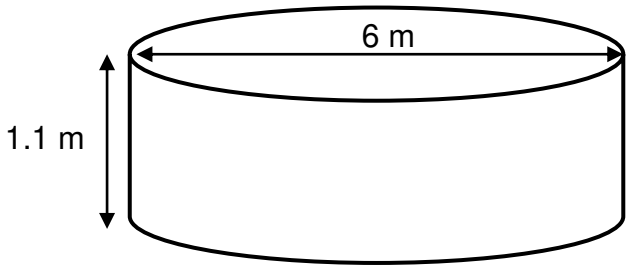
| | |
|---|---|
| **Answer** | **£** |

**[4]**

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**14.** Taylor has a fishpond in the shape of a cylinder.




Not drawn to scale

Taylor uses this formula to work out how many fish can be put into the pond.

*Maximum number of fish = $\frac{\text{volume (in m}^3\text{) of pond multiplied by 220}}{1000}$*

What is the maximum number of fish Taylor can put in the pond?

*Use $\pi = 3.14$*

| **Answer** | **fish** |
|---|---|

**[5]**

**Questions continue on the following page**

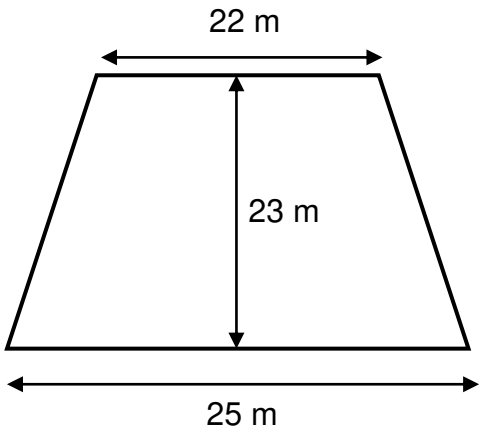
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**15.** Amory has turned a barn into an event venue.

Amory wants to estimate how much money the venue can be rented out for.

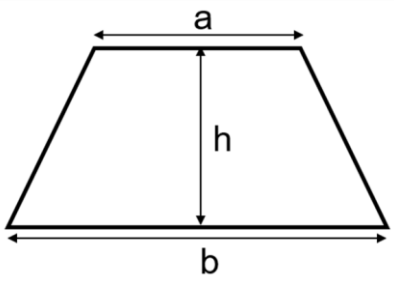
The floor plan of the venue is shown below:




Not drawn to scale

*Use*:

$$Area = \frac{(a+b)h}{2}$$




Amory estimates that one person will require 2.6 $m^2$ of floor space.

Amory finds these prices other local venues charge.

| Venue | A | B | C | D | E |
|---|---|---|---|---|---|
| Price per person | £20 | £35 | £45 | £30 | £50 |

Amory will charge the median price.

What is the maximum amount of money Amory can expect to make?

**Show your working.**

**Answer box is on the following page**

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| | |
|---|---|
| **Answer** | **£** |

**[6]**

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**This is the end of the assessment.**

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